Fizika

9 sinf

Mavzu: **Masalalar yechish**



$$U = \frac{3}{2} \nu\, R\, T = \frac{3}{2}\, p\, V$$

$$A = p_1 \Delta V = \nu\, R\, T$$

$$Q = C\, \Delta T = c\, m\, \Delta T$$

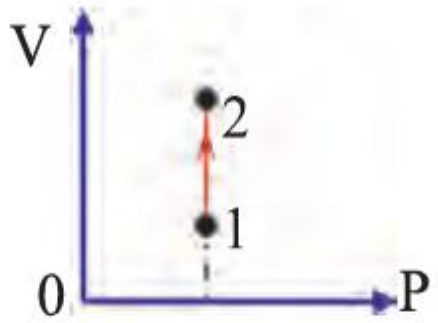
Issiqlik balansi tenglamasi

Issiqlik almashinishi natijasida ichki energiyalari kamaygan (temperaturasi pasaygan) jismlarning uzatgan issiqlik miqdorlarining yig‘indisi ichki energiyasi ortgan (temperaturasi oshgan) jismlarning qabul qilgan issiqlik miqdorlari yig‘indisiga tengdir.

$$Q_1 + Q_2 + Q_3 + \cdots Q_n = Q_1' + Q_2' + Q_3' + \cdots Q_n'$$

Issiqlik almashinuvi jarayonida issiqlik yo‘qdan bor bo‘lmaydi, bordan yo‘q bo‘lmaydi. Issiqlik faqat bir jismdan boshqa jismga o‘tadi.

75-bet, 4-test



Grafikda tasvirlangan jarayonda ideal gazning ichki energiyasi qanday o‘zgaradi?

A) kamayadi; B) ortadi; C) o‘zgarmaydi;

D) avval ortib, so‘ngra kamayadi.

Yechish: $U = \frac{3}{2}PV$ dan

Javob: B) ortadi

75-bet 7-test

1. Moddaning issiqlik sig‘imi quyidagi parametrlarning qaysi biriga bog‘liq?

A) Issiqlik miqdoriga;
B) Modda massasiga;
C) Boshlang‘ich temperaturasiga;
D) Modda turiga.

To‘g‘ri javob: D) modda turiga.

75-bet 13-test



Rasmda bo‘yalgan yuzaning fizik ma’nosi nimadan iborat?

A) bajarilgan ish;
B) temperatura o‘zgarishi;
C) bosimning o‘zgarishi;
D) ma’nosi yo‘q.

Javob: A) bajarilgan ishga teng.

15-mashq 1-masala

Temperaturasi $27^{\circ}C$ va ichki energiyasi 50 kJ bo‘lgan geliy gazining massasi qancha?

Berilgan:

U= 50 kJ= 50000 J

$T = 27 + 273 = 300\ K$

$M = 4\frac{g}{mol} = 0{,}004\frac{kg}{mol}$

$R = 8{,}3\ \frac{J}{K \cdot mol}$

Topish kerak: $m-?$

Formula:

$$U = \frac{3}{2}\ \nu\ R\ T$$

$$\nu = \frac{m}{M}$$

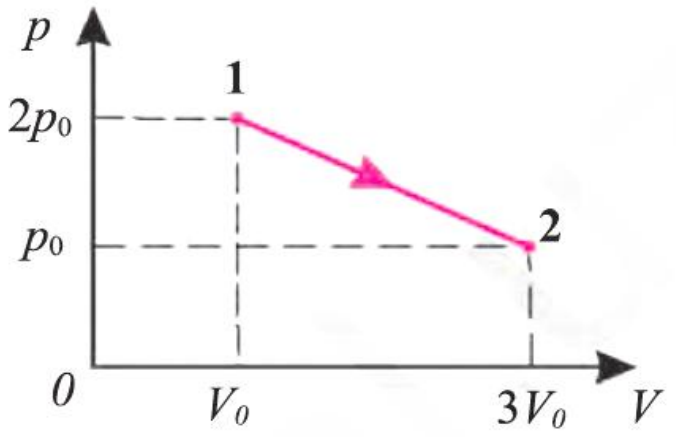
$$U = \frac{3}{2}\ \frac{m}{M}\ R\ T$$

$$m = \frac{2\ M\ U}{3\ R\ T}$$

Yechish: $m = \frac{2 \cdot 0{,}004\ \frac{kg}{mol} \cdot 50000\ J}{3 \cdot 8{,}3\ \frac{J}{K \cdot mol} \cdot 300\ K} \approx 0{,}053\ \text{kg} = 53\ \text{g}$

Javob: $m = 53\ g$

15-mashq, 3-masala



O‘zgarmas bosimli bir atomli ideal gaz 1-holatdan 2-holatga o‘tdi. Bunda uning ichki energiyasi qanday o‘zgaradi?

Gazning dastlabki bosimi $p_0 = 150$ kPa va hajmi $V_0 = 4\ l$ bo‘lgan.

Berilgan:

$p_1 = 2p_0 \qquad V_1 = V_0$

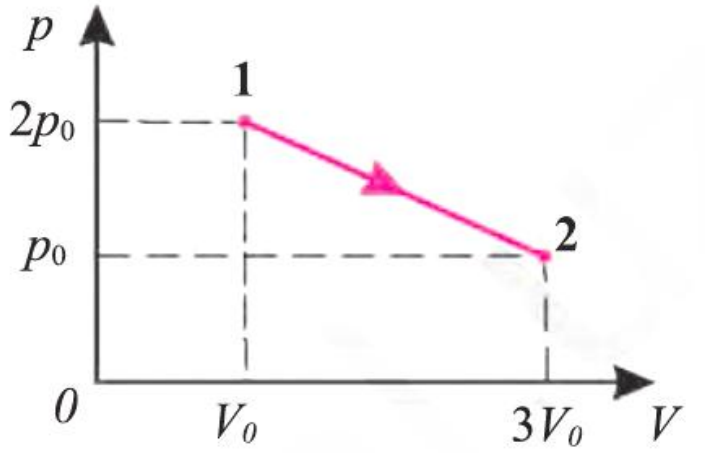
$p_2 = p_0 \qquad V_2 = 3 \cdot V_0$

$V_0 = 4\ l = 0{,}004\ m^3$

$p_0 = 150\ kPa = 150000\ Pa$

Topish kerak: $\Delta U - ?$

Formula:



$U = \frac{3}{2} p\,V$ dan $U_1 = \frac{3}{2} p_1 V_1 = \frac{3}{2} \cdot 2\ p_0 V_0$

$U_1 = 3\ p_0 V_0$ $U_2 = \frac{3}{2}\ p_2 V_2 = \frac{3}{2}\ p_0 \cdot 3\ V_0$

$$U_2 = \frac{9}{2} \cdot p_0 V_0$$

$$\Delta U = U_2 - U_1 = \frac{9}{2}\, p_0 V_0 - 3\, p_0 V_0$$

$$\Delta U = \left(\frac{9}{2} - 3\right) p_0 V_0$$

Yechish: $\Delta U = \left(\frac{9}{2} - 3\right) \cdot 150000\ Pa \cdot 0{,}004\ m^3 =$

$= 900$ J

Javob: 900 J ga ortadi.

15-mashq, 7-masala

8 kg va 90℃ temperaturaga ega bo‘lgan suvga 20℃ temperaturadagi suvdan qancha qo‘shganda, aralashma temperaturasi 30℃ ga teng bo‘ladi?

Berilgan:

$m_1 = 8\ kg$

$t_1 = 90℃$

$t_2 = 20℃$

$t_a = 30℃$

T.K: $m_2 - ?$

Formula:

$$Q_1 = c\, m_1 \Delta t_1 \qquad Q_2 = c\, m_2 \Delta t_2$$

$$\Delta t_1 = (t_1 - t_a) \qquad \Delta t_2 = (t_a - t_2)$$

$$Q_1 = Q_2$$

$$c\, m_1 (t_1 - t_a) = c\, m_2 (t_a - t_2)$$

$$m_1 (t_1 - t_a) = m_2 (t_a - t_2)$$

$$m_2 = \frac{m_1 (t_1 - t_a)}{(t_a - t_2)}$$

Yechish: $m_2 = \frac{8\ kg \cdot (90℃ - 30℃)}{30℃ - 20℃} = 48\ kg$

Javob: $m_2 = 48\ kg$

Mustaqil bajarish uchun topshiriqlar.

15-mashq 4-,5-,9-,10-masalalarni yechish.